R Laboratory - Session 1

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Advanced Statistics for Physics Analysis (2025-2026)

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Additional comments explaining the code in a more detailed way are not reported here, but are available on the corresponding .R source code

We are given a dataset of integer-valued observations $D = \{x_1, x_2, ..., x_N\}$ representing the number of events observed in identical time intervals. The following points will analyse the data using the Bayesian framework.

(1)

We assume that the data is generated from a Poisson distribution, $x_i \sim Poisson(\lambda)$, with a Gamma prior on λ, $\lambda \sim Gamma(\alpha, \beta)$.

(1.a)

To find the posterior distribution $p(\lambda|D)$ we start form the proportional form of the Bayes theorem, $p(\lambda|D) \propto p(D|\lambda) \times p(\lambda)$.

Based on our previous assumption, we have that the likelihood of the data D is :

$$p(D|\lambda) = \prod_{i=1}^{N} \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} \propto \lambda^{\sum_{i=1}^{N} x_i} e^{-N\lambda}$$

and the prior is :

$$p(\lambda) \propto \lambda^{\alpha-1} e^{-\beta\lambda}$$

Therefore, using Bayes theorem we get :

$$p(\lambda|D) \propto p(D|\lambda)p(\lambda) \propto \lambda^{(\sum_{i=1}^{N} x_i + \alpha)-1} e^{-(\beta+N)\lambda}$$

We recognise that this is the form of a Gamma distribution and can rewrite the posterior distribution as :

$$\boxed{p(\lambda|D) = Gamma(\alpha_{\text{post}}, \beta_{\text{post}})} \text{ with } \alpha_{\text{post}} = \alpha + \sum_{i=1}^{N} x_i \text{ and } \beta_{\text{post}} = \beta + N.$$

(1.b)

The Maximum A Posteriori (MAP) estimate is, for a $Gamma(\alpha, \beta)$ distribution, $\frac{\alpha-1}{\beta}$.

$$\lambda_{\text{MAP}} = \frac{\alpha_{\text{post}} - 1}{\beta_{\text{post}}} = \frac{\alpha + \sum_{i=1}^{N} x_i - 1}{\beta + N}$$

Using R, with weakly-informative priors $\alpha = 1, \beta = 0.01$, we can compute λ_{MAP} :

```
lambda MAP estimate : 5.339893
```

(1.c)

To discuss the symmetry of the posterior, we can compute the skewness of the Gamma distribution, given by $\frac{2}{\sqrt{\alpha_{\text{post}}}}$, with the shape parameter, $\alpha_{\text{post}} = \alpha + \sum_{i=1}^{N} x_i$. We then get :

$$\text{Skewness} = \frac{2}{\sqrt{2671}} \approx 0.0386984$$

Because the skewness is extremely close to 0, we can conclude that the posterior distribution is highly symmetric and approximately Gaussian, which is consistent with the Central Limit Theorem.

As we confirmed the symmetry of the posterior distribution, we can use its standard deviation as an estimate of the uncertainty on λ :

$$\sigma_\lambda = \sqrt{\frac{\alpha_{\text{post}}}{\beta_{\text{post}}^2}} = \frac{\sqrt{\alpha_{\text{post}}}}{\beta_{\text{post}}}$$

$$\sigma_\lambda \approx 0.1033614$$

(2)

(2.a)

To derive the posterior predictive distribution for a new observation x_{new}, we marginalize over our parameter λ using the posterior distribution we found in (1.a) :

$$p(x_{new}|D) = \int_0^\infty p(x_{new}|\lambda)p(\lambda|D)d\lambda$$

Substituting the Poisson likelihood and our Gamma posterior :

$$p(x_{new}|D) = \int_0^\infty \left(\frac{\lambda^{x_{new}} e^{-\lambda}}{x_{new}!}\right)\left(\frac{\beta_{\text{post}}^{\alpha_{\text{post}}}}{\Gamma(\alpha_{\text{post}})}\lambda^{\alpha_{\text{post}}-1}e^{-\beta_{\text{post}}\lambda}\right)d\lambda$$

By rearranging the terms and removing the constants from the integral, we get a new Gamma kernel inside :

$$p(x_{new}|D) = \frac{\Gamma(x_{new} + \alpha_{\text{post}})}{x_{new}!\Gamma(\alpha_{\text{post}})}\left(\frac{\beta_{\text{post}}}{\beta_{\text{post}} + 1}\right)^{\alpha_{\text{post}}}\left(\frac{1}{\beta_{\text{post}} + 1}\right)^{x_{new}}$$

We recognize this mathematical form as a Negative Binomial distribution :

$$\boxed{x_{new} \sim NegBin\left(\alpha_{\text{post}}, \frac{\beta_{\text{post}}}{\beta_{\text{post}}+1}\right)}$$

(2.b)

We want to compute the probability $P(x_{new} > 2\overline{x})$. Using R, we first calculate the empirical average $\overline{x}$ of our dataset. Since the predictive Negative Binomial is defined on integers, $P(X > \text{threshold}) = 1 - P(X \leq \lfloor\text{threshold}\rfloor)$. We compute this with the `pnbinom` function :

```r
mean_data <- sum(data) / length(data)
threshold <- 2 * mean_data
prob_success <- beta_post / (beta_post + 1)
prob_exceed <- 1 - pnbinom(q = floor(threshold), size = alpha_post, prob = prob
    _success)
```

```
P(x_new > 2 * mean) : 0.0211426
```

(3)

We now consider an alternative Negative Binomial model : $x_i \sim NegBin(r, \pi)$.

(3.1)

The likelihood is :

$$p(D|r,\pi) = \prod_{i=1}^{N} \frac{\Gamma(x_i + r)}{x_i!\Gamma(r)} \pi^r (1-\pi)^{x_i}$$

For the prior distributions, we choose independent and non-informative priors to let the data drive the inference. Since π is a probability ($\in [0,1]$), we assign a Uniform prior : $p(\pi) \propto 1$. For the parameter $r > 0$, we apply a standard Jeffreys prior : $p(r) \propto \frac{1}{r}$. The joint posterior is therefore proportional to the likelihood times the priors :

$$p(r,\pi|D) \propto \left(\prod_{i=1}^{N} \frac{\Gamma(x_i + r)}{x_i!\Gamma(r)} \pi^r (1-\pi)^{x_i} \right) \frac{1}{r}$$

(3.2)

As the posterior cannot be evaluated analytically, we assume a Gaussian shape and use the Laplace approximation. We optimized the negative log-posterior directly in the original parameter space using the `optim` function in R with the constrained `L-BFGS-B` method to enforce $r > 0$ and $\pi \in (0,1)$.

```r
neg_log_posterior <- function(params, data) {
  r <- params[1]; pi <- params[2]
  ll <- sum(dnbinom(data, size = r, prob = pi, log = TRUE))
  log_prior_r <- -log(r)
  log_prior_pi <- 0
  return(-(ll + log_prior_r + log_prior_pi))
}

opt <- optim(par = c(r_init, pi_init), fn = neg_log_posterior, data = data,
             method = "L-BFGS-B", lower = c(0.001, 0.001), upper = c(Inf, 0.999),
             hessian = TRUE)
cov_matrix <- solve(opt$hessian)
```

The MAP estimates and their associated standard errors (derived from the inverse Hessian matrix in the original space) are :

```
 MAP of r : 7.352101
 MAP of pi : 0.5792658
 Errors (Standard deviations on original parameters) :
[1] 1.13726548 0.03820521
```

(3.3)

Under the Gaussian assumption, we compute the model evidence using the Laplace approximation formula :

$$\log Z \approx -\text{min_nll} + \frac{d}{2}\log(2\pi) - \frac{1}{2}\log|H|$$

```r
d <- 2 # number of parameters (r, pi)
log_det_H <- determinant(opt$hessian, logarithm = TRUE)$modulus
log_evidence_NB <- -opt$value + (d/2)*log(2*pi) - 0.5*log_det_H
```

```
Log-evidence for the negative binomial model : -1237.028
```

(3.4)

To compare the models, we compute the Bayes factor

$$BF = \frac{p(D|M_{NB})}{p(D|M_{Poisson})}$$

Due to computational limits with small probabilities, we evaluate the exact evidence of the Poisson model on a logarithmic scale and compute the log Bayes Factor :

$$\ln(BF) = \ln(p(D|M_{NB})) - \ln(p(D|M_{Poisson}))$$

```r
log_ev_pois_exact <- sum(-lgamma(data + 1)) + alpha_prior*log(beta_prior) -
                     lgamma(alpha_prior) + lgamma(alpha_post) - alpha_post*log(
    beta_post)
log_BF <- log_evidence_NB - log_ev_pois_exact
```

```
Log-evidence for Poisson model : -1279.501
Log Bayes factor (ln BF) : 42.47228
```

(4)

By looking at the results from the previous section, the log Bayes factor is positive and extremely large, meaning the Bayes Factor ($e^{\ln BF}$) largely favours the Negative Binomial model over the Poisson model.

This result is expected. Indeed, the strict Poisson model relies on a strong assumption which is that the mean of the distribution is equal to its variance. However, it can happen to observe overdispersion with real data, where the variance is significantly larger than the mean. The Negative Binomial model introduces the parameter r to add flexibility and capture this overdispersion.

Figure 1 shows the empirical dataset with the Poisson model and the MAP distribution of the Negative Binomial model. We can see that the Poisson distribution is too narrow, whereas the Negative Binomial distribution fits the empirical distribution better.

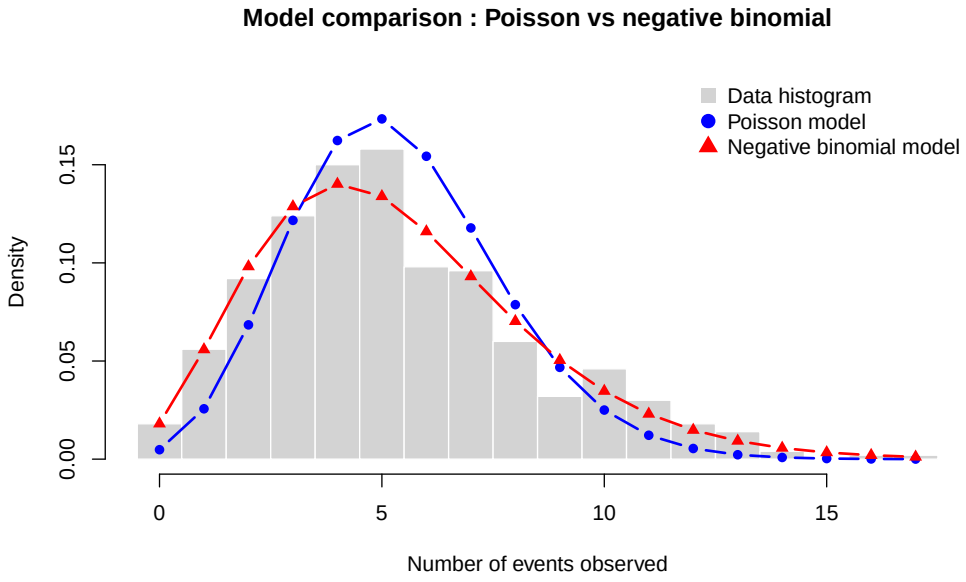

Figure 1: Empirical data compared to Poisson and NegBin MAP models